Idea Submission Template for AI Unlocked

Name of your Institute: *Indian Institute of Management- Calcutta*
Your Team Name: *CATions*
Challenge Area: *AI Study Buddy*
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Ink2IQ AI

From Scribbles to Smart Study

The Team

We are a team of five people with diverse backgrounds. We have 3 BBAs,a BTech and a M Tech. What brings us together is not only our shared mother tongue, but also our enthusiasm for problem solving.As we prepare to begin internships in the field of Quant, Consulting, Marketing and General Management, we are consistently motivated by our desire to understand complex situations and solve the challenges we encounter in our professional journeys and personal lives. During one of our discussions, we were talking about the AI tools we frequently use in our MBA life to help us understand our daily reads and cases with our notes. Through this, we recognised that there is no single tool that is good at assimilating our notes, comprehensively analysing the uploaded case materials and helping us with exam preparation. This realization was our motivation to participate in this challenge. For us, as problem solvers, this is yet another opportunity to take on a meaningful problem and work toward resolving it.

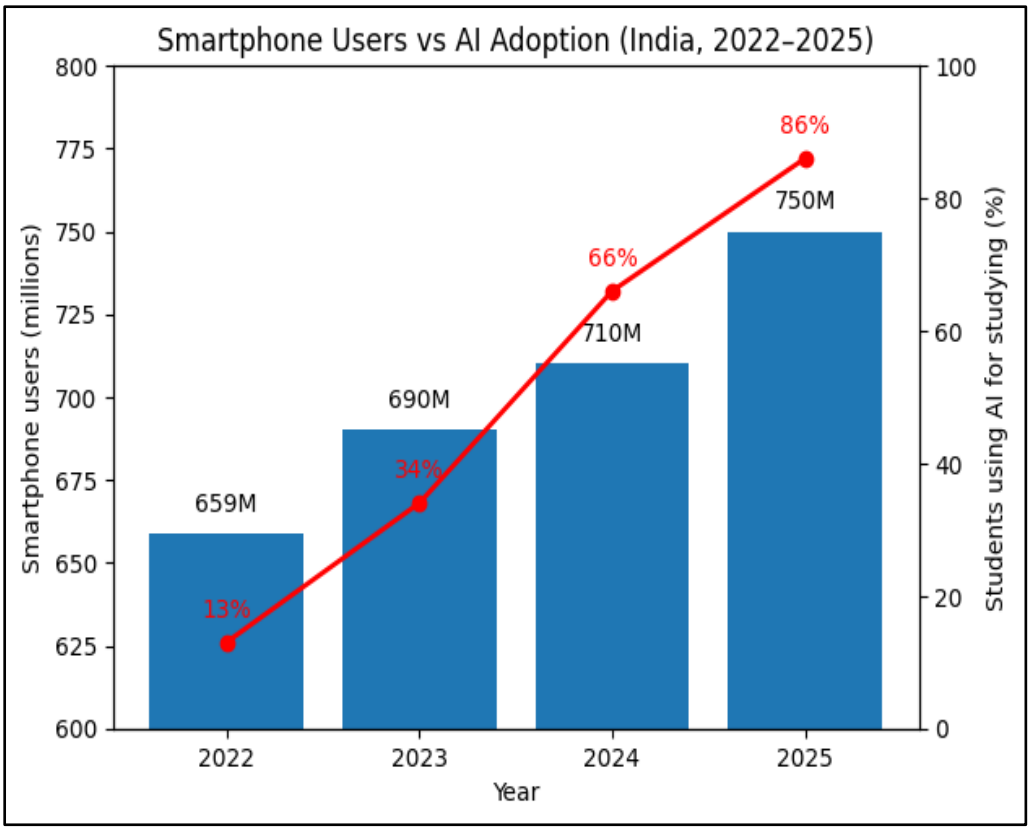
The Concept

Preparing for the UPSC Civil Services Examination is an intense, long-term commitment that demands deep conceptual clarity, consistent revision, and the ability to connect vast amounts of information across subjects like polity, history, economics, and current affairs. Aspirants often rely heavily on handwritten notes, yet revision becomes overwhelming due to information overload, trying to revise through vast notes and personal feedback; especially for working professionals balancing jobs alongside preparation. Our project addresses this problem by building a multimodal AI-powered study companion that understands messy handwritten notes, converts them into structured text, reads them aloud, simplifies complex concepts, and actively quizzes users in both English and their preferred Indian language. By combining image recognition, natural language processing, and speech interaction, the app transforms passive note-taking into active learning. Instead of rereading the same pages repeatedly, aspirants can engage in dynamic revision sessions, concept breakdowns, and personalized testing

Target Audience or Market

Our prime target segment will be UPSC aspirants who are also simultaneously engaged in a full-time job. While the platform will be open to candidates across the country, we will be focussing more on tier 2 & 3 cities as this population is characterised by high smartphone penetration, but lower access to premium coaching facilities. This population also possesses moderate English proficiency and a stronger reliance on bilingual learning.

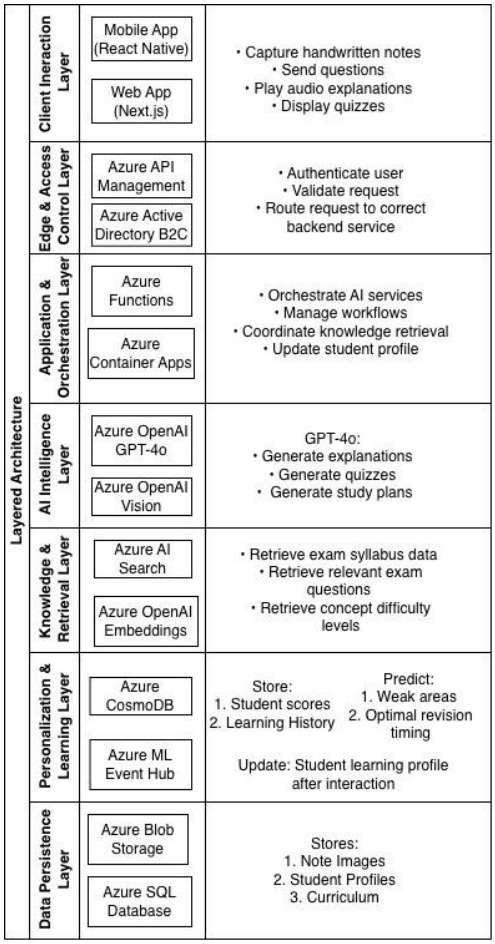
India's competitive-exam prep market is huge and growing rapidly. According to the India Brand Equity Foundation, the country's edtech market is projected to grow from ~$750M in 2020 to $4 billion by 2025 . Millions of students prepare for exams every year: about 1–2 million apply for the UPSC CSE each year , and on the order of 10 million appear in JEE/UPSC-level exams annually . State PSC exams likewise draw hundreds of thousands to millions of aspirants annually (for example, 1.6 million candidates appeared for TNPSC Group IV in 2024 , and 0.58 million for TNPSC Group II). In total, tens of millions of adults nationwide need serious exam preparation, making competitive coaching a multi-billion-dollar opportunity

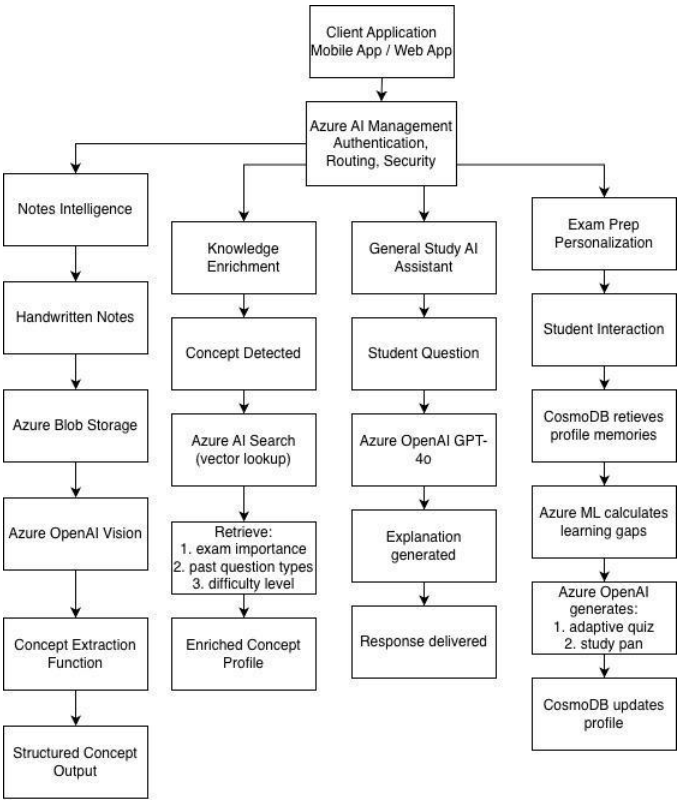


Personas

Persona Attributes	**Arjun(17M)**	**Aditi(22F)**
Education Stage	Class 12	Graduate
Location	Tier-1 city	Tier-2 City
Preparation Type	Mostly coaching-driven (offline/online)	Self-study along with test series
Study Workflow	Lecture → notes → problem solving → revision	Reading → notes → revision → test series → answer writing
Primary Study Material & Notes Type	Coaching modules, PYQs, short notes, formula sheets, tricks, solved examples, short notes	NCERTs, standard books, newspapers, current affairs, long notes, layered notes, mind maps, current affairs snippets
Handwriting Consistency	Medium - low (fast writing)	Medium - high (more structured writing)
Language and notations used	English notes, graphs, circuit diagrams, quick steps	Tamil notes with flowcharts, timelines, tables, case studies
Pain Points with Notes	Notes scattered, hard to revise fast, forget where formula is	Notes become too bulky, hard to revise before exam, hard to link topics
Device Used	Smartphone and Tablet	Smartphone and laptop

How it Works **Core Technologies**





The Business Plan:

Core Multimodal Tutor: Converts messy handwritten notes into structured, syllabus-tagged concepts.
Educational OCR & Auto-Tagging: Extracts formulas, headings, diagrams, and examples while mapping them to exam topics and difficulty levels.
Socratic Concept Teaching: Delivers progressive, step-by-step explanations with guided questioning to correct misconceptions.
Adaptive Assessment Engine: Generates personalized MCQs, short answers, and drills calibrated to learner performance.
Persistent Learner Model: Tracks topic-level mastery, predicts forgetting curves, and optimizes study sequencing.
Knowledge Graph Mapping: Connects syllabus, subtopics, PYQs, and performance gaps into a structured intelligence layer.
Multilingual Speech Interface: Enables seamless English and Indian language learning through integrated ASR and TTS.
Exam-Specific Modules: Provides prelims, mains, essay frameworks, CSAT practice, and state-aligned mock simulations.
Dynamic Revision Planner: Builds exam-date-driven revision schedules with milestone nudges and progress goals.
Coach & Institute Dashboard: Offers cohort analytics, performance heatmaps, bulk uploads, and rubric-based evaluation tools.

Competitive Moat

Capability	Generic LLM	Vision-Language Model (VLM)	RAG-Tutor Bot	Our AI Study Buddy (Integrated Stack)
Handwritten note understanding	Limited (text only)	Reads images but not education-structured	No	Education-tuned OCR + semantic structuring
Diagram & formula extraction	Weak	Partial	No	Structured math/diagram parsing
Syllabus mapping	No structure	No	Retrieval only	Knowledge graph aligned to UPSC/State PSC
Adaptive difficulty	Stateless	No	Limited	Yes (KT + LLM hybrid)
Persistent Learning Module	No	No	No	Yes (concept-level mastery engine)
Socratic step-by-step tutoring	Prompt-dependent	No	Limited	Designed pedagogical layer
Exam-specific intelligence	General	No	Limited	Structured syllabus + PYQ intelligence
Long-term revision planning	No	No	No	Dynamic revision planner
Data flywheel potential	API usage only	Limited	Limited	High (notes + behavior + outcomes)

Go-To-Market (GTM)

Beachhead (0–6 months)

- Product: English MVP (UPSC prelims)
- Users: 500–1,000 invited beta users (topper network + mid-tier coaching partners)
- KPIs: scan→structured accuracy ≥80%; week-4 retention ≥40%; mock score uplift target 15–25% in pilot cohort.

Acquisition channels (high ROI sequencing)

1. Influencer-driven demos: subject experts & micro-creators who do "scan my notes" streams.
2. Communities: Telegram/WhatsApp study groups (time-bound campaigns pre-prelims).
3. SEO/content hub: weekly current affairs + model answers (organic funnel).
4. Institutional pilots: 20 mid-tier coaching centres for white-label trials.

Conversion mechanics

- Free tool hook: 5 free scans + 1 mock.
- Behavioral prompts: "Fix 3 mistakes and unlock a week of Pro" (drive small wins).
- Referral: invite fellow aspirant → both get extra scans.

Scale (6–18 months)

- Regional rollouts by language (Tamil → Marathi → Kannada → Hindi states).
- B2B sales team targeting regional coaching chains (annual contracts).
- Product-led growth: embed shareable progress certificates and "mock leaderboard" (virality).

Operational GTM metrics to track

- CAC by channel, free→paid conversion, DAU/MAU, mock uplift (%), average scans/user, partner retention.